

Resolvent, Scattering, and Heat Dynamics of a One-Dimensional Delta Interaction

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Abstract

We give a convention-complete solution of the one-center delta interaction on the real line in units $\hbar = 2m = 1$. Its pole produces exactly one attractive bound state; the remaining spectrum, left-right scattering, and odd/even channels are closed for every real coupling, including threshold and free faces. Laplace inversion gives the heat kernel and an integrable relative diagonal with explicit trace. Independent exact and high-precision evidence audits all constants without replacing the analytic proof. The model has a natural self-adjoint quantization but no arithmetic-orbit or target-determinant bridge.

1 The frozen singular Hamiltonian

For $\alpha \in \mathbb{R}$, consider the form

$$q_\alpha[\psi] = \int_{\mathbb{R}} |\psi'(x)|^2 dx + \alpha |\psi(0)|^2, \quad \mathcal{D}(q_\alpha) = H^1(\mathbb{R}). \quad (1)$$

The one-dimensional trace inequality makes point evaluation infinitesimally form bounded relative to the Dirichlet integral. Thus (1) is closed and lower bounded and defines one self-adjoint operator H_α . Integration by parts identifies its domain:

$$\mathcal{D}(H_\alpha) = \{\psi \in H^2(\mathbb{R} \setminus \{0\}) : \psi(0+) = \psi(0-), \quad \psi'(0+) - \psi'(0-) = \alpha \psi(0)\}, \quad (2)$$

and $H_\alpha \psi = -\psi''$ off zero. This normalization is the owner; changing the jump sign exchanges attractive and repulsive chambers.

Point interactions and their complete solvability are classical. The monograph of Albeverio, Gesztesy, Høegh-Krohn, and Holden contains a direct chapter on the one-center one-dimensional delta interaction [1]. Our contribution is the self-contained convention and executable closure below, not a priority claim.

2 One scalar equation gives the resolvent

For $\kappa > 0$, the free kernel at energy $-\kappa^2$ is

$$G_\kappa(x, y) = \frac{e^{-\kappa|x-y|}}{2\kappa}.$$

Theorem 1 (Rank-one resolvent). *If $2\kappa + \alpha \neq 0$, then*

$$(H_\alpha + \kappa^2)^{-1}(x, y) = \frac{1}{2\kappa} \left\{ e^{-\kappa|x-y|} - \frac{\alpha}{2\kappa + \alpha} e^{-\kappa(|x|+|y|)} \right\}. \quad (3)$$

Proof. Away from $x = y, 0$, both exponentials solve the homogeneous equation. The free term is continuous and its derivative has jump -1 at $x = y$. At zero, the kernel value in (3) is $e^{-\kappa|y|}/(2\kappa + \alpha)$. The image term has derivative jump $\alpha e^{-\kappa|y|}/(2\kappa + \alpha)$, exactly α times that value. It therefore satisfies (2) and the resolvent source equation. Uniqueness follows from invertibility away from the displayed pole. $\square$

The formula is a Krein rank-one correction. The exceptional equation $2\kappa + \alpha = 0$ has a positive solution precisely when $\alpha < 0$; it is spectral data, not a missing regular grid point.

3 Spectrum and two-channel scattering

The odd subspace automatically vanishes at zero, so H_α is free there. On the even subspace the interface becomes the half-line Robin condition $\psi'(0+) = (\alpha/2)\psi(0)$. The odd sine transform and the even Robin generalized eigenfunctions

$$\varphi_{\alpha,k}(x) = \sqrt{\frac{2}{\pi}} \frac{2k \cos(kx) + \alpha \sin(kx)}{\sqrt{4k^2 + \alpha^2}}, \quad x > 0, \quad k > 0,$$

give the two continuum transforms; when $\alpha < 0$, the even transform is completed by the single normalized bound vector in (4).

Theorem 2 (Complete sign atlas). *The essential spectrum is $[0, \infty)$ and is purely absolutely continuous; the singular-continuous spectrum is empty. If $\alpha < 0$, there is exactly one simple eigenvalue and normalized eigenfunction,*

$$E_b = -\frac{\alpha^2}{4}, \quad \phi_b(x) = \sqrt{-\frac{\alpha}{2}} e^{\alpha|x|/2}. \quad (4)$$

For $\alpha \geq 0$ there is no eigenvalue.

For momentum $k > 0$, a unit-amplitude wave incident from the left has

$$r_\alpha(k) = \frac{\alpha}{2ik - \alpha}, \quad t_\alpha(k) = \frac{2ik}{2ik - \alpha}. \quad (5)$$

The scattering matrix has odd eigenvalue 1 and even eigenvalue $(2ik + \alpha)/(2ik - \alpha)$, both of modulus one.

Proof. The resolvent pole gives (4); direct integration gives $\|\phi_b\|_2^2 = (-\alpha/2)(-2/\alpha) = 1$. Stone's formula applied to the explicit odd Dirichlet and even Robin resolvents yields the Lebesgue-density transforms displayed above. Their denominator $4k^2 + \alpha^2$ has no positive-real zero, the zero-energy solution is not in L^2 , and the only off-continuum singularity is the displayed attractive pole. Hence there is no embedded point or singular-continuous spectrum. For scattering, write $e^{ikx} + re^{-ikx}$ on $x < 0$ and te^{ikx} on $x > 0$. Continuity gives $t = 1 + r$ and the jump gives $2ikr = \alpha t$, proving (5). Finally

$$|r|^2 = \frac{\alpha^2}{4k^2 + \alpha^2}, \quad |t|^2 = \frac{4k^2}{4k^2 + \alpha^2},$$

so flux is one; diagonalizing the symmetric left-right matrix gives the two channel phases. $\square$

At $\alpha = 0$ the operator is exactly free. For nonzero coupling, the zero-energy limit is perfectly reflecting and the high-energy limit is transparent. The attractive pole occurs at $k = i|\alpha|/2$; a repulsive coupling has no upper-half-plane pole.

4 Heat dynamics and the relative trace

Let $K_0(t; u) = (4\pi t)^{-1/2} e^{-u^2/(4t)}$. Laplace inversion of the two terms in (3) uses, for $\Re\sqrt{s} > \max\{0, -\alpha/2\}$, the one-line identity

$$\mathcal{L}_{t \rightarrow s} \left[e^{\alpha a/2 + \alpha^2 t/4} \operatorname{erfc} \left(\frac{a}{2\sqrt{t}} + \frac{\alpha\sqrt{t}}{2} \right) \right] = \frac{2e^{-a\sqrt{s}}}{\sqrt{s}(2\sqrt{s} + \alpha)}.$$

It yields the following all-time expression.

Theorem 3 (Heat kernel and trace defect). *For $t > 0$, $x, y \in \mathbb{R}$, and $a = |x| + |y|$,*

$$K_\alpha(t; x, y) = K_0(t; x - y) - \frac{\alpha}{4} e^{\alpha a/2 + \alpha^2 t/4} \operatorname{erfc} \left(\frac{a}{2\sqrt{t}} + \frac{\alpha\sqrt{t}}{2} \right). \quad (6)$$

The diagonal difference from the free heat kernel is integrable and

$$\int_{\mathbb{R}} \{K_\alpha(t; x, x) - K_0(t; 0)\} dx = \frac{1}{2} \left[e^{\alpha^2 t/4} \operatorname{erfc} \left(\frac{\alpha\sqrt{t}}{2} \right) - 1 \right]. \quad (7)$$

Proof. The Laplace transform in t of (6) is (3); uniqueness of the semigroup transform proves the kernel. For the trace, symmetry reduces the image term to

$$-\frac{\alpha}{2}e^{\alpha^2 t/4} \int_0^\infty e^{\alpha x} \operatorname{erfc}\left(\frac{x}{\sqrt{t}} + \frac{\alpha\sqrt{t}}{2}\right) dx.$$

One integration by parts evaluates the integral as

$$\frac{e^{-\alpha^2 t/4} - \operatorname{erfc}(\alpha\sqrt{t}/2)}{\alpha},$$

with its continuous value at $\alpha = 0$, giving (7). □

The trace defect tends to zero as $t \downarrow 0$. For $\alpha < 0$, its large-time leading term is $e^{\alpha^2 t/4}$, the heat weight of the unique negative eigenvalue. This is not growth of the unitary group e^{-itH_α} , which remains norm preserving. For $\alpha > 0$, the relative trace tends to $-1/2$.

5 Executable closure and Route-A boundary

The producer contains 32 regular resolvent cells, three deliberately separated poles, 28 scattering cells, three bound-state cells, and eight high-precision heat cells. A strict duplicate-rejecting checker reports 1,726 assertions and reconstructs the heat values by inverse Laplace transformation and an integrated diagonal resolvent; a symbolic engine reports 46 identities; two fresh outputs are byte identical; and 30/30 semantic, structural, raw-JSON, and stale-hash attacks fail as required. These finite receipts audit signs and factors; they do not prove Theorems 1–3.

The natural self-adjoint Hamiltonian earns only `A4_NATURAL_QUANTIZATION`. No rational-prime carrier, primitive repetition law, logarithmic arithmetic clock, target divisor, target functional equation, or Hilbert–Pólya operator is constructed. Under the literal scope `NO_BAD_EULER_OR_ROOT_NUMBER`, the strict tuple is

$$(\text{A0_FAIL}, \text{A1_FAIL}, \text{A2_FAIL}, \\ \text{A3_FAIL}, \text{A4_NATURAL_QUANTIZATION}),$$

the overall verdict is `ROUTE_A_REJECTED`, and Route B is disabled.

References

- [1] S. Albeverio, F. Gesztesy, R. Høegh-Krohn, and H. Holden, *Solvable Models in Quantum Mechanics*, Theoretical and Mathematical Physics, Springer, 1988, [doi:10.1007/978-3-642-88201-2](https://doi.org/10.1007/978-3-642-88201-2).